

**How the Public Utilities Regulatory Commissions Act Influenced
the Effect of Utility-Scale Solar as a Climate Solution**

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Introduction

Climate change describes the rapid evolution of natural landscapes as resources are depleted due to human activities. This phenomenon impacts local, regional, and global climates and is accelerated by the combustion of fossil fuels, which emphasizes the planet's greenhouse effect. Using ground, air, and space technologies to measure these changes, NASA observed a positive correlation between emissions from fossil fuels and rising average surface temperatures (n.d.). Burning nonrenewable energy sources like coal and natural gas for energy emits heat-trapping particles in excessive amounts, making the planet's atmospheric "greenhouse" work in excess. Other key indicators of global warming include glacier melt and ice loss around Earth's polar regions, intensified weather patterns and more frequent extreme events like hurricanes, and variance in vegetation cover. Lindsey & Dahlman (2025) studied how human activities have influenced global warming patterns since the establishment of the United Nations Framework Convention on Climate Change in 1994. They observed an amplified amount of warming in the past 30 years compared to the previous century with figures illustrating hot spots in the same regions for both time scales, further demonstrating the permanence of climate change.

While there are many approaches to address this phenomenon, creating relevant policies that outline large-scale implementation is highly beneficial as they ensure decisions made by governments, businesses, and individuals are cohesive. Long-term goals of such entities guide the policy making process by providing context for what is to be done, who is to do it, how it will be completed, and for whom it will affect (Gifford, 2025a). The US energy sector has been largely influenced by greenhouse gas policies that aim to mitigate climate change impacts with regulatory, economic, and informational tools. Specifically, utility-scale solar technologies have emerged as a climate solution with potential to save 42.32 gigatons of carbon dioxide emissions in a 2C temperature increase scenario and 119.13 gigatons of carbon dioxide emissions in a 1.5C temperature increase scenario (The Carbon Almanac Network, 2022). These systems leverage the abundant and inexhaustible sun to produce energy that can power buildings, cars, and everything in between.

Much of this energy transition can be attributed to the National Energy Act of 1978 which promoted energy conservation, grid independence, and renewable sourcing through several provisions that encouraged solar (Howard, 1980). The National Energy Conservation Policy Act focused on informing residential users of renewable energy benefits and increasing access to solar while the Energy Tax Act generated federal credits for homeowners and businesses that installed certain technologies. On the other hand, the Powerplant and Industrial Fuel Use Act prohibited utilities from burning natural gas and oil unless combined with alternate fuels. Arguably the most important statute within this 1978 legislation is the Public Utilities Regulatory Policies Act (PURPA) which detailed energy retail rates and allowed smaller producers to connect to utility transmission lines. This supported decentralized power production by reassuring

users of financial stability. A fifth provision, The Natural Gas Policy Act, is included in the 1978 decision but indirectly impacts solar technologies.

The Public Utilities Regulatory Policies Act

Implementation of PURPA followed a top-down approach as Congress mandated that electric utilities across the country purchase some of their energy from cogeneration facilities and small power production facilities (APPA, n.d.). Cogeneration facilities recycle the waste heat produced in traditional power generation and repurpose it for heating water or buildings, resulting in a 75% operational energy efficiency (Industrial Technologies Office, n.d.). Small power production facilities are primarily supplied by renewable, biomass, waste, or geothermal sources with working capacities of 80 megawatts or less (FERC, 2025). The Federal Energy Regulatory Commission oversees the identification of qualifying facilities (QF) eligible to participate in this mandate based on efficiency standards set by the agency. While this climate policy was created at the national level, it was effectively scaled down to the state level where their Public Utility Commissions determined avoided costs of alternative purchases, labeled Investment Tax Credit (ITC), and reviewed utility/QF contract conditions.

PURPA has been celebrated as one of the first pieces of clean-energy legislation that catalyzed the energy transition and discouraged fossil fuel use. Stabilizing purchase rates allowed renewable energy projects like utility-scale solar to be financially viable and competitive for the first time in US history. With the creation of QF standards, energy conservation became a reality as more efficient production methods were explored with financial incentives to achieve energy savings. Shortly after the National Energy Act was established in 1978, the Atlantic Richfield Company opened a 1.1-megawatt, utility-scale solar power plant in Hesperia, California (SEIA Comms Team, 2024). A year later, Minnesota introduced a net metering program to compensate residential and commercial customers for their solar energy exported back to the regional grid (DSIRE, n.d.). This portfolio diversification enhanced Former President Jimmy Carter's goal to decrease foreign energy dependence by creating a new market commodity. Businesses, homeowners, and organizations alike turned their attention to deploying solar technologies across the US which pressured utilities to re-evaluate their fossil fuel energy projects if they wanted to stay relevant.

With these outcomes in mind, PURPA would generally be considered successful in its objective to support decentralized and domestic power production. Through federal regulation of utility energy sourcing and the market progression for renewables, solar technologies have greatly improved in efficiency and accessibility. The four other statutes no doubt helped to develop the solar industry but this provision has paved the way for future climate policies. For example, the Energy Policy Act of 2005 pioneered a new perspective for valuing renewable energy sources as it raised the ITC from 10% to 30% of a system's initial cost while introducing the Production Tax Credit (PTC) that offered annual compensation for each kilowatt-hour produced (Lips, 2024). These two

mechanisms were once again updated in 2022 with Former President Joe Biden's Inflation Reduction Act which expanded the criteria for ITC expensing, cutting the average residential rooftop solar costs by approximately 7,500 2022 USD (SETO, 2022). Policies like the Energy Policy Act of 2005 and Inflation Reduction Act of 2022 have further promoted solar technologies and the nation's original 1978 goals. States are also organizing their own programs like Renewable Portfolio Standards and Interconnection Standards to encourage in-state solar development as a way for electricity suppliers to meet their obligations (EPA, n.d.).

Policy Gaps & Challenges

While PURPA provided many benefits, this 1978 statute was not perfect. Since state PUCs had the discrepancy to set avoided cost rates for QFs, large inconsistencies resulted between states and within actual market rates (Brindle, 2019). Forecasting fixed costs led some PUCs to generate long-term contracts at highly optimistic rates that assumed more load growth and higher fuel prices when the opposite occurred. With renewable energy sources becoming cheaper than ever, these contracts have severely impacted the effectiveness of PURPA as ratepayers take on more burden. This is the case in North Carolina where a significant number of solar facilities were QF eligible, leading consumers to pay more during the contract's duration (NC Department of Justice, 2019).

As a whole, PURPA was not designed to properly account for modern systems as the renewable energy market recognized the risks of intermittency, carbon financing emerged as another climate intervention, and grid needs rapidly changed with technological development. Because solar intensities are influenced by the time of day, season, and current weather conditions, energy production can greatly vary with the grid being under or over supplied (Pitra & Sastry Musti, 2021). Updated climate goals reference activity emissions that contribute to global warming which PURPA incidentally addresses through renewables. Mechanisms for reductions focus on capacity building and technology transfers such as carbon markets and emissions trading (Gifford, 2025b) which has potential to impact solar technology investment. Similarly, the development of wholesale electricity markets and open access transmission make regional grids more complex as new entities were able to join and drive innovation (Kavulla & Murphy, 2018).

The current policy landscape reflects another challenge as the federal administration plans to restrict legislation built from PURPA's foundation. President Donald Trump's One Big Beautiful Bill Act (OBBB) was signed this past year and marks a halt in the renewable energy industry expansion as fossil fuel energy production is prioritized. This legislation minimizes the Inflation Reduction Act of 2022's timeline for ITC eligibility and coerces projects to start sooner with completion deadlines (Klymenko, 2025). Although OBBB does not directly change PURPA's legal status, it does reduce the incentive for developers to explore solar and elicit higher contract

prices. With less financial motivation, utility-scale solar technologies are less affordable and therefore less effective as a climate solution.

Proposal

Given the gaps outlined in the prior section, the Public Utilities Regulatory Policies Act of 1978 has a few different methods for evolving to better fit modern needs. Despite the uncertainty around the future of renewables, I would recommend a combination of the following practices to advance utility-scale solar technologies and strengthen PURPA as a statute.

First and foremost, drafting national standards to determine avoided cost rates would reduce conflict and ease integration. This practice is key to successful project implementation across the nation because a consistent set of guidelines helps regional developers understand how/where/when they can interact with the climate policy. It simplifies the issue at hand by offering a series of steps that are compatible with a variety of scenarios. Additional benefits of standardization include faster project deployment, more grid reliability, and policy reinforcement when used as a regulatory tool (Sustainability Director, 2025). Avoided cost rates should also incorporate mechanisms that adjust market prices according to the environmental impact of the activity since key indicators of climate change have become more stressed. Carbon financing is a popular strategy for emission reductions that can increase the accuracy of avoided cost rates as it reflects how consumers discount the future. Shareholders can then support initiatives with more assurance that their money will be returned with less risk for “future-proof investing” (Green.Earth, 2025).

Intermittency is another important challenge that must be tackled for PURPA to remain effective. Rewriting the criteria for QF eligibility to include energy storage facilities and protect cogeneration or small power production from utility discrimination. Battery energy storage systems bridge the gap between supply and demand for intermittent sources as they provide an easy way to hold and dispatch energy at various scales (WSP, 2025). This buffer of sorts proactively works to reduce energy load during demand fluctuations and conserve supplies so energy is conserved. Establishing interconnection standards would further enhance grid capacities as QF are guaranteed a fair and efficient appraisal process for utility-scale solar technologies (State Climate Policy Dashboard, n.d.). These standards keep renewable sources competitive in the market as they lower project risk due to increased predictability of what each developer must do to be eligible. Collaboration and cooperation between all entities is crucial throughout the entire reformation of PURPA. The four proposed ideas for modernizing this statute are most effective when paired together but are unfortunately not plausible given the upcoming changes to national environmental programs. However, it is still worth evaluating how policy, like PURPA, can help fortify solutions that mitigate the impacts of climate change.

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